

1. Introduction

Organizations increasingly rely on quantitative analysis to guide workforce decisions, particularly in compensation, hiring, and performance management. This project analyzes the TechSales_Reps dataset, which contains demographic, educational, performance, and personality information for sales representatives. The objective is to develop a regression model to predict employees' salaries in the software product group, identify the most influential predictors, and translate these insights into actionable managerial recommendations. Model performance was evaluated using RMSE, MAE, and R^2 on both training and validation (holdout and k-fold cross-validation with $k=10$) datasets to ensure generalizability.

2. Data Description, Quality Assessment, and Preprocessing

The dataset contains 21,990 observations and includes variables such as Age, Years of experience, Certificates, College, Gender, Personality, NPS, and Salary. Because the assignment requires focusing on the software division, the dataset was filtered to include only employees with Business = "Software", resulting in 12,130 observations.

Data Quality

Initial inspection revealed three rows with blank Business values and empty factor levels in College and Personality.

After cleaning, no missing values remained in the variables used for modelling. Descriptive statistics for key variables are reported below.

Variable	Mean	SD	Min	Max
Salary	72176.751855	22628.964787	21000	192000
Age	39.004617	10.363314	21	57
Years	2.423825	2.165357	1	13
NPS	6.299423	2.177552	1	10

These key statistics confirm substantial variation in salary, age, years, and NPS, supporting the need for more flexible modelling. No extreme outliers were removed, as they appeared to reflect genuine variation.

Preprocessing

Categorical variables were recoded into binary indicators (Female_Yes, College_Yes). Personality was encoded using three dummy variables (Pers_Analyst, Pers_Diplomat, Pers_Sentinel), with Explorer as the baseline. Explorer was chosen as the baseline because it is the largest personality group. The Feedback variable was removed. The dataset was

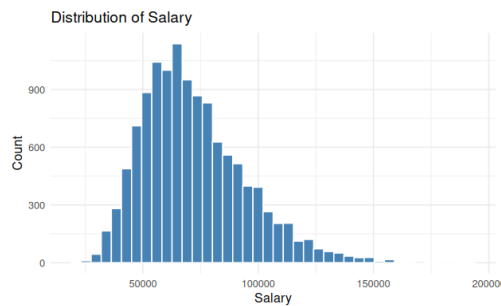
split into 70% for training and 30% for validation to support out-of-sample evaluation. Standard mean-centring of continuous predictors was used to account for potential multicollinearity and to estimate interaction terms.

3. Exploratory Data Analysis (EDA)

EDA was conducted to understand variable distributions and guide model specification.

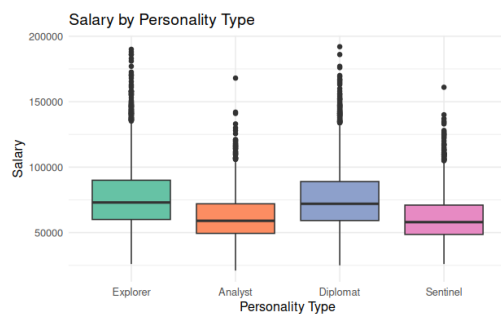
Salary Distribution

Salary was strongly right-skewed, motivating the later use of a log-transformation.



Salary by Personality

Boxplots revealed clear differences across personality types: Analysts and Sentinels earned less on average than Explorers and Diplomats.



Other Relationships (visuals omitted for space)

NPS showed a positive association with Salary, with slight curvature. A correlation matrix confirmed moderate correlations among some variables, but no multicollinearity severe enough to prevent joint inclusion. No other significant trends were found in the data.

4. Model Development

The modelling process progressed through six increasingly refined specifications. The baseline linear model revealed clear curvature and heteroscedasticity, prompting the addition of quadratic terms in Model 2, which improved fit but

did not address the skewness in salary. Model 3 resolved much of this issue by modelling $\log(\text{Salary})$, producing more stable residuals and a clearer interpretation. Models 4 and 5 introduced interaction terms—first directly, then with centred predictors—but these interactions were not significant and did not improve adjusted R^2 . The final model, Model 6, retained the log-salary structure and included nonlinear terms for Age and NPS along with all relevant main effects, offering the best balance of interpretability, parsimony, and predictive performance. MAE was reported but emphasized less because RMSE is more sensitive to large errors, aligns directly with the OLS loss function used to estimate the models, and provides clearer differentiation between competing specifications during cross-validation.

Model 6:

$$\text{Log}(\text{Salary}) = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{Age}^2) + \beta_3(\text{Years}) + \beta_4(\text{NPS}) + \beta_5(\text{NPS}^2) + \beta_6(\text{Certificates}) + \beta_7(\text{Female_Yes}) + \beta_8(\text{College_Yes}) + \beta_9(\text{Pers_Analyst}) + \beta_{10}(\text{Pers_Diplomat}) + \beta_{11}(\text{Pers_Sentinel}) + \varepsilon$$



Residual plots for all Models

5. Model Comparison

Across the six models, predictive accuracy improves as a nonlinear structure is added. The baseline linear model performs the weakest, with the highest training RMSE and lowest R^2 . Introducing polynomial terms into the nonlinear model reduces error and increases explained variance, indicating that salary exhibits meaningful curvature with respect to Age, Years, and NPS. The log-salary and interaction models offer no improvement over the nonlinear specification. The centred nonlinear interaction model achieves the best in-sample performance, but the interaction effects themselves add little value. The final model matches the best training performance while remaining more parsimonious, and it also achieves the lowest cross-validation RMSE, indicating the strongest generalization. CV RMSE is the most reliable metric because it evaluates performance across multiple folds, reducing sensitivity to a single train/validation split. Overall, the final model provides the best balance of accuracy and interpretability.

Model	Train_RMSE	Train_MAE	Train_R2	CV_RMSE	Validation_RMSE
Linear	15944.78	12273.54	0.5031969	15959	15859.25
Nonlinear	15533.66	11934.41	0.5284856	15552	15439.81
Log Salary	15890.82	12036.82	0.5065533	15207	15779.92
Interaction	15890.72	12037.17	0.5065599	15208	15780.45
Centered Interaction	15369.59	11599.83	0.5383931	14692	15246.46
Final Model	15370.17	11600.95	0.5383585	14690	15245.08

6. Interpretation of the Final Model

The final model indicates that salary is strongly related to Age and NPS in a nonlinear way: earnings rise with both variables, but at diminishing rates. Years of experience also has a positive effect, and holding additional certificates is associated with a modest but statistically significant increase in salary. Demographic factors are essential as well—women earn less than comparable men, while having a college degree is associated with substantially higher pay. All three personality types (Analyst, Diplomat, and Sentinel) have negative coefficients relative to the baseline Explorer group, indicating that, holding other factors constant, these personalities tend to earn less than Explorers. However, we cannot statistically determine whether Diplomats differ from Explorers, as the coefficient is not significant.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.481e+00	3.821e-02	248.142	< 2e-16 ***
Age	5.234e-02	1.777e-03	29.454	< 2e-16 ***
I(Age^2)	-5.559e-04	2.262e-05	-24.576	< 2e-16 ***
Years	3.082e-03	1.040e-03	2.962	0.00306 **
NPS	5.703e-02	6.218e-03	9.171	< 2e-16 ***
I(NPS^2)	-1.429e-03	4.730e-04	-3.020	0.00253 **
Certificates	6.222e-02	1.538e-03	40.448	< 2e-16 ***
Female_Yes	-8.357e-02	4.433e-03	-18.853	< 2e-16 ***
College_Yes	1.532e-01	5.651e-03	27.106	< 2e-16 ***
Pers_Analyst	-1.385e-01	7.631e-03	-18.152	< 2e-16 ***
Pers_Diplomat	-5.639e-03	5.150e-03	-1.095	0.27354
Pers_Sentinel	-1.489e-01	7.078e-03	-21.038	< 2e-16 ***

Interpreting coefficients on the log scale, a one-unit increase in Age increases salary by approximately 5.2% at the margin, before diminishing due to the negative quadratic term. A 10-point increase in NPS corresponds to roughly a 5.7% increase in salary, again with diminishing returns. Women earn about 8.4% less than comparable men, while college graduates earn approximately 15.3% more. Analysts and Sentinels earn 13.8% and 14.9% less than Explorers, respectively. Although some coefficients are small in magnitude, their effects accumulate meaningfully across the salary distribution, especially for Age, NPS and College. Each additional year of experience increases salary by approximately 0.3% in the final model. Earlier specifications included a squared Year term that suggested diminishing returns to experience. Still, this effect was not statistically significant and did not reduce cross-validation RMSE, so only the linear term was retained for the final model. At the sample mean salary of \$72,177, one additional year of age corresponds to roughly \$3,750 in additional compensation. Similarly, the college premium implies an average increase

of approximately \$11,000, while the gender gap corresponds to roughly \$6,100. The intercept represents the expected log-salary for a baseline employee (Explorer, male, non-college, zero certificates) at Age=0 and NPS=0, which is not meaningful but serves as a model anchor.

7. Managerial Recommendations

The final model suggests that compensation is strongly tied to Age, NPS, and experience, highlighting the value of retention and customer-satisfaction initiatives. Supporting employees in earning additional certificates may offer a cost-effective boost to productivity. Gender and education gaps warrant review to ensure equitable pay practices. Because Analysts, Diplomats, and Sentinels earn less than Explorers even after controlling for performance-related factors, managers should examine whether role assignments or evaluation criteria unintentionally favour certain personality profiles. Aligning compensation more closely with measurable performance rather than personality traits may improve fairness and retention. HR can deploy the final model as a salary benchmarking and equity-monitoring tool. Employees whose actual salaries deviate substantially from their predicted values could be flagged for review, enabling proactive detection of pay inequalities or misaligned compensation.

8. Limitations and Future Work

Because this analysis uses observational data, causal conclusions cannot be drawn. Important predictors, such as territory size, sales quotas, or product mix, were unavailable and may have contributed to additional salary variation. Future work could incorporate more complete datasets and more complex models (e.g., tree-based models) to capture more complex patterns. Periodic retraining would be required to maintain accuracy as workforce characteristics evolve.

9. Conclusion

This analysis shows that salary variation in software sales is driven by nonlinear effects of Age and NPS, as well as experience, education, gender, certificates, and personality. Comparing six models demonstrated that incorporating curvature and using a log-salary transformation yields the best balance of accuracy and interpretability. The final model generalizes well and highlights meaningful compensation patterns, including gender gaps and personality-related disparities. These insights provide a defensible foundation for compensation strategy, pay-equity review, and targeted workforce development.